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Research Interests: Industrial Organization, Generative AI, Microeconomics
Creative Works

Education

Ph.D. in Economics, University of Toronto — 2026 (Expected)
Committee: Matthew Mitchell (co-supervisor), Heski Bar-Isaac (co-supervisor)
Avi Goldfarb, Ambarish Chandra

M.A in Economics, University of Toronto — 2020

B.A in Economics, Fudan University — 2019

Research

Human Effort In Production With GenAI (JMP)

Hiding From Generative AI, R&R at Quantitative Economics

Platforms in Platform

Awards and Grants

TD MDAL Grant, Rotman School of Management, University of Toronto	2025
TD MDAL Grant, Rotman School of Management, University of Toronto	2024
University of Toronto Doctoral Fellowship	2020 - 2025

Conference Presentations

TSE Digital Economics Conference (Toulouse)	2025
North American Summer Meeting of Econometric Society (Nashville)	2024
ESIF Economics and AI+ML Meeting of Econometric Society (Ithaca)	2024
European Economic Association Annual Meeting (Rotterdam)	2024
American Law and Economics Annual Meeting (Ann Arbor)	2024
Canadian Economics Association Annual Meeting (Toronto)	2024

Computer Skills

Python, STATA, MATLAB, LaTeX, Cloud Computing

References

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Abstracts

Human Effort In Production With GenAI

(Job Market Paper)

How important is human effort in production with generative AI? This paper investigates users' production process of visual arts with an influential AI image generator, Midjourney. Using prompt-level interaction history data of artists across multiple AI version releases, including both prompts submitted to AI and resulting images, I find that more recent AI versions produce outputs that more closely match the prompts and are more preferred by artists. Text analysis of prompts reveals that artists change prompts when AI improves. By submitting prompts from old AI to new AI, output image shifts can be decomposed into effects from prompt changes and effects from AI development. This counterfactual experiment shows that over 70% of output shifts come from prompt changes, instead of AI capability gains. I also document novel facts of prompt patterns in image production and find that they are similar to consumer product searching patterns. (1) Prompt length increases step-by-step; (2) Prompts are path dependent;(3) Prompts converge to the final prompt along the paths; (4) Artists adjust words with greater weights first and then move to less important words. I interpret these findings through a sequential search framework in Weitzman (1979), and estimate a structural model of prompt construction. The counterfactual investigates when part of human judgment is removed from prompt construction, how does it impact the final production results.

Hiding From Generative AI

How does generative Artificial Intelligence (AI) change the behaviors of content creators? I investigate the effect of an AI image generator on artists' incentives to publish artworks using data from an online art platform, DeviantArt. On November 11 2022, DeviantArt introduced a generative AI image generator into the platform and artworks on this platform entered training data by default. Using a difference-in-differences estimation with artists who do not use AI, I show that digital artists publish 21% fewer artworks following AI's introduction on this platform, in contrast to artisan crafts artists. This reduction could potentially hinder knowledge spillovers to other artists and AI training data availability. By matching the artworks of artists who publish both on DeviantArt and Instagram, I find that despite artists publishing fewer artworks on DeviantArt, the quality of published artworks for a given artist remains the same after the introduction of AI.

Platforms in Platform

Platforms could be part of a larger platform like an app store. Even though there exists a positive cross-group network externality between buyers and sellers, there is still an incentive for buyers and sellers to segment into different platforms. Buyers may choose to use different platforms to access more targeted products, while sellers may use different platforms to avoid competition for buyers' attention. This paper argues that segmenting the market using smaller platforms can be profitable for the large platform, as it can charge different platform fees to sellers of different quality and save buyers' attention to better sellers. This paper also shows that when market agglomeration is more profitable for the large platform, it may choose to ban smaller platforms, which could be socially inefficient as it may deter mediocre sellers from entering the market.